

Non-Intrusive Reduced-Order Modeling for Viscoelastic Extrudate Swell

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Abstract

This work introduces a non-intrusive reduced-order model (ROM) framework that combines proper orthogonal decomposition (POD) with Gaussian process regression (GPR). The framework is applied to two-dimensional planar and three-dimensional cylindrical extrudate swell configurations. An adaptive sampling strategy is also developed to select training snapshots sequentially by maximizing the GPR predictive uncertainty, and it is compared to space-filling Latin hypercube sampling. The numerical results demonstrate that the ROM can accurately capture the steady-state extrudate shape and the full transient swelling dynamics over the parameter space at a fraction of the cost of full-order simulations.

Keywords: Extrudate swell, Viscoelastic fluid, Giesekus model, Reduced-order modeling, Proper orthogonal decomposition, Gaussian process regression, Adaptive sampling, Non-intrusive surrogate

1. Introduction

Viscoelastic fluids are encountered in many natural and industrial processes, *e.g.* polymer extrusion, additive manufacturing, coating flows, and biological transport phenomena. A feature of these materials is that they can store elastic energy upon deformation and release it upon the removal of geometric constraints. When a viscoelastic fluid exits a die, the confinement is suddenly removed, and the elastic stresses can relax, causing an expansion of the free surface, often referred to as *extrudate swell*. The amount of expansion depends on the rheological properties of the material and the flow conditions inside the die [1, 2, 3]. Therefore, extrudate swell is observed in a wide range of extrusion technologies, including polymer melt processing, food extrusion, extrusion-based additive manufacturing, 3D printing, and pharmaceutical melt extrusion [4, 5, 6, 7, 8].

Accurate and efficient simulation of extrudate swell remains a challenging task due to the strong coupling between viscoelastic stress evolution and free-surface deformation, as well as the stress singularity that arises

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at the die exit. Capturing this interaction requires resolving steep stress gradients that develop near the die exit while simultaneously tracking the evolving interface of the extrudate. Resolving these features requires a highly refined spatial discretization, particularly in the vicinity of the die exit, where stress gradients are highly nonlinear. In many finite-element formulations, the moving interface is handled using an Arbitrary Lagrangian–Eulerian (ALE) framework, while the nonlinear and often stiff nature of viscoelastic constitutive equations further restricts the allowable time-step size. These difficulties become even more pronounced in three-dimensional simulations [9, 10, 11]. Consequently, applications that require repeated simulations over a range of material parameters or different flow conditions, such as process optimization, uncertainty quantification, inverse material characterization, or real-time control, can become very expensive when relying directly on the full-order model (FOM). This computational burden motivates the development of surrogate and reduced-order modeling strategies capable of reproducing the essential flow physics at a significantly lower computational cost.

Reduced-order modeling (ROM) addresses this challenge by seeking a low-dimensional subspace to approximate the high-fidelity solutions [12, 13]. ROM strategies can be broadly classified into two types: intrusive and non-intrusive [14]. Intrusive methods require the projection of the governing equations onto a reduced basis, for which hyper-reduction techniques, such as the Discrete Empirical Interpolation Method (DEIM) [15, 16], are then needed to efficiently evaluate the remaining nonlinear terms in *online* queries. Intrusive approaches can be very accurate, but they require direct access to and modification of the full-order solver and can develop stability issues in the strongly nonlinear regime characteristic of high-Weissenberg-number viscoelastic flows [17, 16]. Non-intrusive methods do not suffer from these issues because they treat the FOM as a black box: a surrogate is trained on snapshot data to map physical parameters to reduced coefficients without any modification of the solver. The surrogate is typically selected as radial basis functions [18], Gaussian process regression (GPR) [19] and neural networks [20].

Reduced-order modeling has been extensively studied for Newtonian fluid dynamics, but its application to viscoelastic flows is relatively limited, with only a few recent works addressing such systems [21, 22, 17]. For these problems, projection-based intrusive ROM approaches can be used, but the strong nonlinearities of viscoelastic constitutive models can lead to a loss of stability. Moreover, most of the existing ROM frameworks assume a fixed computational domain, while the extrudate-swell problem involves an evolving free surface and a deforming computational mesh, usually treated within an ALE formulation. To the authors’ knowledge, reduced-order modeling has not been applied to the viscoelastic extrudate-swell problem that couples viscoelastic stress evolution and free-surface dynamics at the same time. The present work aims to

fill this gap in the literature.

We propose a non-intrusive reduced-order modeling framework designed for the viscoelastic extrudate-swell problem. The method combines Proper Orthogonal Decomposition (POD), which extracts a low-dimensional spatial basis from high-fidelity simulation snapshots, with Gaussian Process Regression (GPR) to learn the mapping between physical parameters and the corresponding POD coefficients. In addition, the probabilistic nature of GPR provides an estimate of the predictive uncertainty, which we exploit to design an adaptive sampling strategy that identifies the most informative parameter locations for additional full-order simulations. This approach reduces the number of required training snapshots while maintaining the desired level of accuracy. The performance of the proposed ROM framework is demonstrated for both two-dimensional planar and three-dimensional cylindrical extrudate-swell configurations governed by the Giesekus constitutive model under steady and transient flow conditions.

The remainder of the paper is organized as follows. Section 2 presents the governing equations, the Giesekus constitutive model, the ALE formulation, and the finite element discretization that constitute the FOM. Section 3 describes the non-intrusive ROM pipeline: snapshot construction, POD basis extraction, GPR surrogate construction, and the adaptive sampling algorithm. Numerical results for the 2D planar and 3D cylindrical configurations are reported and discussed in Section 4. Conclusions and perspectives for future work are drawn in Section 5.

2. Full-order model

The full-order model (FOM) is used to generate the high-fidelity snapshot data for the reduced-order model. It resolves the coupled viscoelastic flow, free-surface deformation, and mesh motion in the extrudate-swell problem. This section summarizes the computational domain, governing equations, constitutive model, ALE formulation, and numerical discretization used in the FOM.

2.1. Problem description

We consider the extrusion of a viscoelastic fluid from a confined die into an unconfined region, as illustrated in Figure 1. Upon exiting the die, the elastic stresses developed during the confined flow relax, causing an expansion of the free surface known as the *extrudate-swell* phenomenon. Two geometrical configurations are investigated: a two-dimensional planar die (Figure 1a) and a three-dimensional cylindrical die (Figure 1b). In both cases, the computational domain consists of a die section of length $L_{\text{die}} = 3$ connected to an extrusion region of length $L_{\text{extr}} = 5$. The die exit is located at $x = 0$, with the die occupying $x \in [-L_{\text{die}}, 0]$

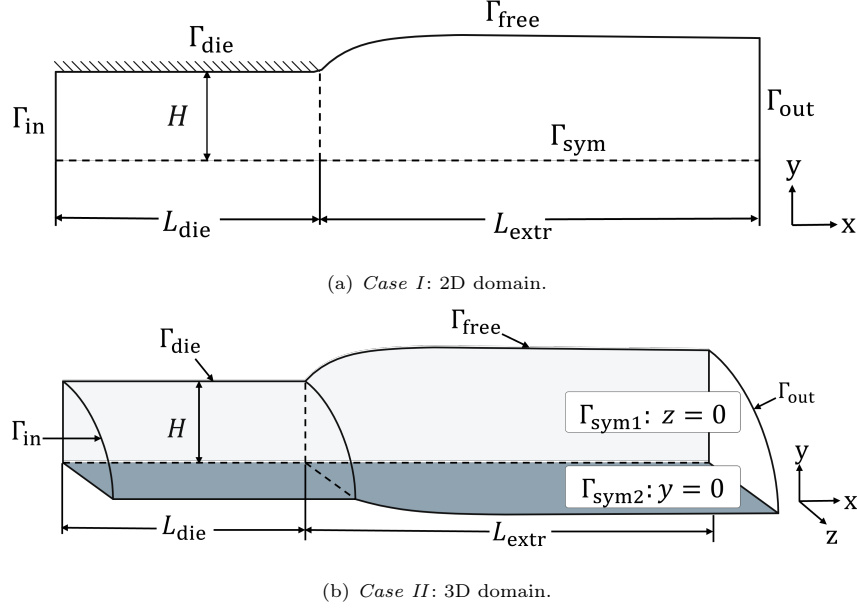


Figure 1: Schematic of the extrudate-swell computational domains.

and the extrusion region occupying $x \in [0, L_{extr}]$. The characteristic die dimension is $H = 1$, corresponding to the die half-height in the planar configuration and the die radius in the cylindrical configuration. Symmetry is exploited to reduce the computational cost. For the planar geometry, only the upper half of the domain is simulated, while for the cylindrical geometry only one quarter of the domain is considered. The fluid enters the domain through the inlet boundary Γ_{in} at a prescribed flow rate Q and exits through the outlet boundary Γ_{out} . The imposed flow rate is $Q = U_{avg}H$ for the planar configuration and $Q = \frac{1}{4}\pi U_{avg}H^2$ for the cylindrical configuration, where U_{avg} denotes the mean axial velocity.

2.2. Balance and constitutive equations

We consider an incompressible flow in the creeping-flow regime, where inertial effects are negligible and no external forces are present. We also neglect the surface tension. Under these assumptions, the governing equations reduce to the conservation of mass and momentum, given by

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ denotes the fluid velocity, and $\boldsymbol{\sigma}$ is the Cauchy stress tensor. The stress tensor is decomposed into three contributions: an isotropic pressure, a viscous solvent stress, and a polymeric extra stress. This

decomposition can be written as

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta_s\mathbf{D} + \boldsymbol{\tau}, \quad (3)$$

where p is the pressure, η_s is the solvent viscosity, and $\boldsymbol{\tau}$ represents the polymeric stress contribution. The rate-of-deformation tensor $\mathbf{D}$ is defined as $\mathbf{D} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$. The polymeric stress tensor $\boldsymbol{\tau}$ is related to the conformation tensor $\mathbf{c}$ by

$$\boldsymbol{\tau} = \frac{\eta_p}{\lambda}(\mathbf{c} - \mathbf{I}) = G(\mathbf{c} - \mathbf{I}), \quad (4)$$

where $G = \frac{\eta_p}{\lambda}$ is the polymer shear modulus, η_p is the polymer viscosity, λ is the relaxation time and $\mathbf{I}$ is the identity tensor. The evolution of the conformation tensor is given by

$$\frac{D\mathbf{c}}{Dt} = \mathbf{L} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{L}^T - f(\mathbf{c}), \quad (5)$$

where $\mathbf{L} = \nabla\mathbf{u}$ is the velocity-gradient tensor, with the convention $L_{ij} = \partial u_i / \partial x_j$. The function $f(\mathbf{c})$ is described by the Giesekus model:

$$f(\mathbf{c}) = \frac{(\mathbf{c} - \mathbf{I}) + \alpha(\mathbf{c} - \mathbf{I})^2}{\lambda}, \quad (6)$$

where λ is the relaxation time, and α is the mobility parameter governing shear-thinning effects.

2.3. Boundary and initial conditions

A schematic representation of the domains is shown in Figure 1. At the inlet Γ_{in} , fully developed inflow conditions are imposed by solving a separate periodic inlet problem. No-slip conditions are applied along the die wall Γ_{die} , while the traction vanishes on the free surface Γ_{free} . At the outlet Γ_{out} , a free outflow condition is prescribed, which means that there is no velocity in the direction transverse to the flow. On the symmetry boundaries, the velocity normal to the boundary is set to zero.

Case I: 2D

The prescribed boundary conditions for the 2D domain are

$$\mathbf{u} = \mathbf{u}_{\text{fd}} \quad \text{on } \Gamma_{\text{in}}, \quad (7)$$

$$\mathbf{c} = \mathbf{c}_{\text{fd}} \quad \text{on } \Gamma_{\text{in}}, \quad (8)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_{\text{die}}, \quad (9)$$

$$u_y = 0 \quad \text{on } \Gamma_{\text{sym}}, \quad (10)$$

$$u_y = 0 \quad \text{on } \Gamma_{\text{out}}, \quad (11)$$

where $\mathbf{u}_{\text{fd}}$ and $\mathbf{c}_{\text{fd}}$ denote the fully developed velocity and conformation profiles corresponding to the imposed flow rate Q . The evolution of the free surface Γ_{free} is governed by the 2D height function $h(x, t)$:

$$\frac{\partial h}{\partial t} + u_x \frac{\partial h}{\partial x} = u_y, \quad x \in [0, L_{\text{extr}}]. \quad (12)$$

The height function is initialized as $h(x, 0) = H$, and $h(0, t) = H$ attaches the free surface to the die exit.

Case II: 3D

The prescribed boundary conditions for the 3D case are

$$\mathbf{u} = \mathbf{u}_{\text{fd}} \quad \text{on } \Gamma_{\text{in}}, \quad (13)$$

$$\mathbf{c} = \mathbf{c}_{\text{fd}} \quad \text{on } \Gamma_{\text{in}}, \quad (14)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_{\text{die}}, \quad (15)$$

$$u_z = 0 \quad \text{on } \Gamma_{\text{sym1}}, \quad (16)$$

$$u_y = 0 \quad \text{on } \Gamma_{\text{sym2}}, \quad (17)$$

$$u_y = u_z = 0 \quad \text{on } \Gamma_{\text{out}}. \quad (18)$$

Let (r, θ, z) denote cylindrical coordinates aligned with the die axis, and let u_r , u_θ , and u_z be the corresponding radial, azimuthal, and axial velocity components. We track the free surface as a radial height function $h(\theta, z, t)$, which evolves according to the kinematic condition [23]

$$\frac{\partial h}{\partial t} + u_z \frac{\partial h}{\partial z} + \frac{u_\theta}{r} \frac{\partial h}{\partial \theta} = u_r, \quad (\theta, z) \in [0, \pi/2] \times [0, L_{\text{extr}}], \quad (19)$$

evaluated at the free surface $r = h(\theta, z, t)$. The surface starts out as a cylinder of radius H , so that $h(\theta, z, 0) = H$, and the condition $h(\theta, 0, t) = H$ keeps it attached to the die exit.

2.4. Numerical methods

The governing equations are discretized using the finite element method. Quadratic (P_2) elements are employed for the velocity field, while linear (P_1) elements are used for both the pressure and the conformation tensor fields, resulting in a P_2 - P_1 - P_1 discretization. To account for the evolution of the free surface, an Arbitrary Lagrangian–Eulerian (ALE) formulation is employed [24, 25]. To enhance numerical stability at moderate and high Weissenberg numbers, several stabilization techniques are employed. These include the log-conformation formulation [26, 27], streamline-upwind/Petrov-Galerkin (SUPG) stabilization [28], and

the DEVSS-G method [29]. Time integration is performed using a second-order backward differentiation scheme (BDF2).

3. Reduced-order model

The reduced-order model (ROM) approximates the parameterized full-order solutions in a low-dimensional space. The framework combines proper orthogonal decomposition (POD) for basis construction with Gaussian process regression (GPR) to predict the reduced coefficients from the input parameters. The *offline* stage consists of assembling snapshot matrices, computing POD bases, and training the GPR surrogate. In the *online* stage, a new parameter query requires only a surrogate evaluation followed by a low-rank reconstruction, at a cost independent of the full-order dimension N_h . The two stages are quantified for the present configurations in Appendix Appendix B.

3.1. Proper orthogonal decomposition

Let $\mathbf{u}(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ denote the vector of nodal values of a high-fidelity solution field for a parameter vector $\boldsymbol{\mu} \in \mathcal{P} \subset \mathbb{R}^d$, where N_h is the total number of degrees of freedom and d is the dimension of the parameter space $\mathcal{P}$. Each $\mathbf{u}(\boldsymbol{\mu})$ collects the nodal values of a field of interest, namely the velocity or the trace of the conformation tensor. For a finite training set $\mathcal{S} = \{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_M\} \subset \mathcal{P}$, the snapshots are assembled column-wise:

$$\mathbf{X} = [\mathbf{u}(\boldsymbol{\mu}_1), \mathbf{u}(\boldsymbol{\mu}_2), \dots, \mathbf{u}(\boldsymbol{\mu}_M)] \in \mathbb{R}^{N_h \times M}. \quad (20)$$

Note that although the extrudate-swell geometry varies with the parameters, all snapshots are represented using meshes with identical topology and degree-of-freedom ordering. Each entry of every snapshot vector corresponds to the same discrete degree of freedom across different parameter realizations. Therefore, the snapshots can be assembled consistently into a snapshot matrix.

The column space of $\mathbf{X}$ provides a discrete approximation of the parameterized solution manifold

$$\mathcal{M}_h = \{\mathbf{u}(\boldsymbol{\mu}) : \boldsymbol{\mu} \in \mathcal{P}\}.$$

Since only finitely many parameter instances can be sampled, the ROM approximates the sampled manifold determined by $\mathcal{S}$. Therefore, its accuracy depends on both the number of snapshots and their distribution within the parameter domain. Standard space-filling strategies, such as Latin hypercube sampling (LHS) or Halton sequences, may be used to construct $\mathcal{S}$. The adaptive sampling strategy introduced in Section 3.4

replaces these fixed designs with an uncertainty-driven approach that selects new high-fidelity samples in the regions where the predictive uncertainty of the surrogate model is highest.

Algorithm 1 POD basis construction.

- 1: **Input:** snapshot matrix $\mathbf{X} \in \mathbb{R}^{N_h \times M}$, POD truncation tolerance ϵ_{pod} .
- 2: Compute the economy singular value decomposition

$$\mathbf{X} = \mathbf{\Phi} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{\Phi} \in \mathbb{R}^{N_h \times M}, \quad \mathbf{V} \in \mathbb{R}^{M \times M}, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_M),$$

with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M \geq 0$.

- 3: Normalize the singular-value spectrum: $\bar{\sigma}_i = \sigma_i / \sigma_1$, $i = 1, \dots, M$.
- 4: Select the reduced dimension $r = \max\{i : \bar{\sigma}_i > \epsilon_{\text{pod}}\}$.
- 5: Form the reduced basis from the leading r left singular vectors:

$$\mathbf{\Phi}_r = \mathbf{\Phi}(:, 1:r) \in \mathbb{R}^{N_h \times r}.$$

- 6: **Output:** reduced basis matrix $\mathbf{\Phi}_r$ and reduced dimension r .
-

POD constructs a low-dimensional orthonormal basis that captures the dominant structures of the snapshot data through the singular value decomposition (SVD) of the snapshot matrix. The snapshot matrix is decomposed as

$$\mathbf{X} = \mathbf{\Phi} \mathbf{\Sigma} \mathbf{V}^\top, \tag{21}$$

where $\mathbf{\Phi} \in \mathbb{R}^{N_h \times M}$ and $\mathbf{V} \in \mathbb{R}^{M \times M}$ have orthonormal columns, and $\mathbf{\Sigma}$ contains the singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$. The columns of $\mathbf{\Phi} = [\phi_1, \phi_2, \dots]$ form an orthonormal basis, and each singular value σ_i quantifies the amount of snapshot energy captured by the associated mode. By the Eckart–Young theorem [30, 31], the best rank- r approximation of $\mathbf{X}$ in the Frobenius norm is obtained by truncation to the leading r modes,

$$\mathbf{X} \approx \mathbf{\Phi}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top, \tag{22}$$

where $\mathbf{\Phi}_r \in \mathbb{R}^{N_h \times r}$ collects the first r left singular vectors. In practice, the truncation rank is determined from the normalized singular value spectrum through a user-prescribed POD truncation tolerance ϵ_{pod} : r is taken as the largest index for which the normalized singular value σ_i / σ_1 exceeds ϵ_{pod} . Throughout this work, the POD truncation tolerance is fixed at $\epsilon_{\text{pod}} = 10^{-6}$ for all reduced-order models. The procedure is summarized in Algorithm 1.

3.2. Solution reconstruction

For any parameter value, the corresponding reduced coefficients $\mathbf{a}(\boldsymbol{\mu}) \in \mathbb{R}^r$ are the coefficients of the solution in the POD basis, obtained by projecting:

$$\mathbf{a}(\boldsymbol{\mu}) = \Phi_r^\top \mathbf{u}(\boldsymbol{\mu}), \quad (23)$$

so that $\mathbf{u}(\boldsymbol{\mu}) \approx \Phi_r \mathbf{a}(\boldsymbol{\mu})$. Evaluating $\mathbf{a}(\boldsymbol{\mu}^*)$ directly for a new parameter $\boldsymbol{\mu}^*$ would, however, still require the full-order solution $\mathbf{u}(\boldsymbol{\mu}^*)$ or a Galerkin projection of the governing equations onto Φ_r . To avoid this, a regression model $\hat{\pi} : \mathbb{R}^d \rightarrow \mathbb{R}^r$ is trained *offline* on the dataset $\mathcal{D} = \{(\boldsymbol{\mu}_i, \mathbf{a}(\boldsymbol{\mu}_i)) : i = 1, \dots, M\}$ to learn the parameter-to-coefficient mapping:

$$\boldsymbol{\mu} \mapsto \mathbf{a}(\boldsymbol{\mu}). \quad (24)$$

Once trained, the ROM prediction for any new $\boldsymbol{\mu}^* \in \mathcal{P}$ is obtained entirely from the surrogate:

$$\mathbf{u}_{\text{rom}}(\boldsymbol{\mu}^*) = \Phi_r \hat{\pi}(\boldsymbol{\mu}^*), \quad (25)$$

bypassing the full-order model during *online* evaluation.

Using the same POD procedure and regression gives a reduced approximation of the moving mesh coordinates for a given parameter. The predicted coordinates reconstruct the swollen mesh on which the predicted field is then defined. As will be shown in Section 4, the normalized singular values of the mesh coordinates decay faster than those of either physical field, so that the geometry is represented by the fewest modes and with the smallest projection error of the three.

In the non-intrusive ROM literature, several regression strategies have been proposed to approximate the map $\boldsymbol{\mu} \mapsto \mathbf{a}(\boldsymbol{\mu})$, such as polynomial regression, radial basis functions, neural networks, and Gaussian process regression. For a small training dataset and a moderate number of parameters, Gaussian process regression (GPR) is a popular choice. Moreover, GPR offers uncertainty estimates on the reconstructed solution, which can be utilized for adaptive sampling of the parameter space.

3.3. Gaussian process regression for reduced-order model

Gaussian process regression (GPR) is a non-parametric, kernel-based probabilistic framework for approximating an unknown regression function $f : \mathcal{P} \rightarrow \mathbb{R}$ from a set of observational data. Let $\{(\boldsymbol{\mu}_i, y_i) : i = 1, \dots, M\}$ denote the training set of M observations, where $\boldsymbol{\mu}_i \in \mathcal{P} \subset \mathbb{R}^d$ is the i -th input parameter vector and $y_i \in \mathbb{R}$

is the corresponding scalar output. The observed input-output pairs are assumed to satisfy

$$y_i = f(\boldsymbol{\mu}_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2), \quad (26)$$

where ε_i is an independent Gaussian noise term with variance σ_n^2 , accounting for numerical or measurement error.

A GP is a probability distribution over functions such that any finite collection of function values is jointly Gaussian. It is fully characterized by its mean function $m(\boldsymbol{\mu}) = \mathbb{E}[f(\boldsymbol{\mu})]$ and covariance function $\kappa(\boldsymbol{\mu}, \boldsymbol{\mu}') = \text{Cov}(f(\boldsymbol{\mu}), f(\boldsymbol{\mu}'))$, and we write

$$f \sim \mathcal{GP}(m(\cdot), \kappa(\cdot, \cdot)). \quad (27)$$

We assume a zero prior mean, so the GP prior is determined entirely by the kernel $\kappa(\cdot, \cdot)$, which encodes assumptions about the smoothness and correlation structure of f in the parameter space.

We adopt the automatic relevance determination (ARD) squared-exponential kernel,

$$\kappa(\boldsymbol{\mu}, \boldsymbol{\mu}') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_{k=1}^d \frac{(\mu_k - \mu'_k)^2}{\ell_k^2}\right), \quad (28)$$

where σ_f^2 denotes the signal variance and ℓ_k are the characteristic length scales that control the sensitivity of the regression model to each parameter direction.

The training dataset is provided by

$$\mathcal{D} = \{(\boldsymbol{\mu}_i, \mathbf{a}(\boldsymbol{\mu}_i))\}_{i=1}^M,$$

with the input matrix

$$\mathbf{P} = \begin{bmatrix} \boldsymbol{\mu}_1^\top \\ \vdots \\ \boldsymbol{\mu}_M^\top \end{bmatrix} \in \mathbb{R}^{M \times d}, \quad \mathbf{Z} = \begin{bmatrix} \mathbf{a}_1 = \mathbf{a}(\boldsymbol{\mu}_1)^\top \\ \vdots \\ \mathbf{a}_M = \mathbf{a}(\boldsymbol{\mu}_M)^\top \end{bmatrix} \in \mathbb{R}^{M \times r}.$$

Since the quantity of interest is the reduced coefficient vector $\mathbf{a}(\boldsymbol{\mu}) \in \mathbb{R}^r$, the regression problem consists of approximating the mapping $\boldsymbol{\mu} \mapsto \mathbf{a}(\boldsymbol{\mu})$. This is achieved by assigning an independent Gaussian process to each coefficient $a_j(\boldsymbol{\mu})$, $j = 1, \dots, r$, while using a common covariance kernel for all outputs.

Let $\mathbf{K} = \mathbf{K}(\mathbf{P}, \mathbf{P})$ denote the kernel matrix,

$$\mathbf{K}_{ij} = \kappa(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j) + \sigma_n^2 \delta_{ij}. \quad (29)$$

Since the snapshots are generated by deterministic full-order simulations, the observations are noise-free.

Therefore, σ_n^2 is not treated as a hyperparameter but is fixed to a small value $\sigma_n^2 = 10^{-10}$.

For a new parameter $\boldsymbol{\mu}^*$, define the cross-covariance vector

$$\mathbf{k}_* = \begin{bmatrix} \kappa(\boldsymbol{\mu}_1, \boldsymbol{\mu}^*) \\ \vdots \\ \kappa(\boldsymbol{\mu}_M, \boldsymbol{\mu}^*) \end{bmatrix}, \quad k_{**} = \kappa(\boldsymbol{\mu}^*, \boldsymbol{\mu}^*).$$

Under the Gaussian process prior, and since the outputs are conditionally independent given the shared kernel, the joint distribution of the j -th column $\mathbf{z}_j \in \mathbb{R}^M$ of $\mathbf{Z}$ and the corresponding coefficient $a_j(\boldsymbol{\mu}^*)$ is Gaussian:

$$\begin{bmatrix} \mathbf{z}_j \\ a_j(\boldsymbol{\mu}^*) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbf{K} & \mathbf{k}_* \\ \mathbf{k}_*^\top & k_{**} \end{bmatrix} \right), \quad j = 1, \dots, r. \quad (30)$$

Conditioning on the observed data yields the posterior predictive distribution

$$\mathbf{a}(\boldsymbol{\mu}^*) \mid \mathbf{Z} \sim \mathcal{N}(\mathbf{Z}^\top \mathbf{K}^{-1} \mathbf{k}_*, (k_{**} - \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{k}_*) \mathbf{I}_r), \quad (31)$$

where the posterior mean provides the predicted reduced coefficients for all modes simultaneously, and the scalar variance term, common to every output, quantifies the predictive uncertainty at $\boldsymbol{\mu}^*$. The kernel hyperparameters $\{\sigma_f^2, \ell_1, \dots, \ell_d\}$ are determined by maximizing the log-marginal likelihood,

$$\mathcal{L} = -\frac{1}{2} \text{tr}(\mathbf{Z}^\top \mathbf{K}^{-1} \mathbf{Z}) - \frac{r}{2} \log \det \mathbf{K} - \frac{Mr}{2} \log(2\pi), \quad (32)$$

which balances data fidelity and model complexity.

The Gaussian process is used as the regression model $\hat{\pi}$ defined in Equation (24). The posterior mean provides the predicted coefficients $\hat{\mathbf{a}}(\boldsymbol{\mu}^*)$ from which the reduced solution $\mathbf{u}_{\text{rom}}(\boldsymbol{\mu}^*) = \Phi_r \hat{\mathbf{a}}(\boldsymbol{\mu}^*)$ is reconstructed. Similarly, the posterior variance is used for adaptive sampling. The complete ROM procedure is summarized in Algorithm 2.

Algorithm 2 POD-based GPR reduced model

offline stage:

- 1: Compute the full-order snapshots and assemble the snapshot matrix $\mathbf{X} \in \mathbb{R}^{N_h \times M}$.
- 2: Perform truncated SVD on $\mathbf{X}$ to obtain the POD basis $\Phi_r \in \mathbb{R}^{N_h \times r}$.
- 3: Project snapshots onto Φ_r to obtain the coefficient matrix $\mathbf{Z} \in \mathbb{R}^{M \times r}$.
- 4: Train a GPR surrogate on $\mathcal{D} = \{(\boldsymbol{\mu}_i, \mathbf{a}(\boldsymbol{\mu}_i)) : i = 1, \dots, M\}$.

online stage:

- 5: For a new parameter $\boldsymbol{\mu}^*$, predict the coefficient vector $\hat{\mathbf{a}}(\boldsymbol{\mu}^*)$ via the trained GPR surrogate.
 - 6: Reconstruct the ROM solution: $\mathbf{u}_{\text{rom}}(\boldsymbol{\mu}^*) = \Phi_r \hat{\mathbf{a}}(\boldsymbol{\mu}^*)$.
-

3.4. Adaptive sampling

The prediction error at unseen parameter values arises from two distinct sources: the POD projection error and the Gaussian process regression (GPR) error associated with approximating the POD coefficients. For a fixed reduced basis, the latter contribution typically dominates and increases in regions of the parameter space that are insufficiently sampled. To mitigate this effect, we employ a *standard-deviation-based adaptive sampling* strategy, following the active learning approach of [19], which iteratively enriches the training set at the locations where the predictive standard deviation of the surrogate is largest.

Starting from an initial training set comprising M parameter–snapshot pairs, the procedure proceeds sequentially. At each iteration, a multi-output GPR model with a shared kernel across all outputs is trained on the dataset

$$\mathcal{D}_M = \{(\boldsymbol{\mu}_i, \mathbf{a}(\boldsymbol{\mu}_i))\}_{i=1}^M,$$

where $\mathbf{a}(\boldsymbol{\mu}_i) \in \mathbb{R}^r$ denotes the vector of POD coefficients associated with the parameter $\boldsymbol{\mu}_i$. A candidate pool $\mathcal{C} = \{\boldsymbol{\mu}_j\}_{j=1}^{N_c}$ is then generated over the parameter domain by Latin hypercube sampling.

Since the kernel is shared across all outputs, the posterior standard deviation is common to every reduced coefficient and is adopted directly as the uncertainty indicator,

$$\xi_{\text{sd}}(\boldsymbol{\mu}) = \hat{\sigma}(\boldsymbol{\mu}), \quad (33)$$

where $\hat{\sigma}$ denotes the posterior standard deviation. The next parameter to be added to the training set is selected as the maximizer of this indicator over the candidate pool:

$$\boldsymbol{\mu}_{M+1} = \arg \max_{\boldsymbol{\mu} \in \mathcal{C}} \xi_{\text{sd}}(\boldsymbol{\mu}). \quad (34)$$

A high-fidelity simulation is then performed at $\boldsymbol{\mu}_{M+1}$, the resulting snapshot is appended to the training

set, and both the POD basis and the GPR surrogate are updated accordingly. The reduced model is thus rebuilt at every iteration rather than merely refined so that the truncation rank r may change from one iteration to the next. The prediction error is evaluated on the test samples at each iteration and recorded for the convergence study of Figure 10. This plays no role in the selection, which depends solely on Eq. (33) and therefore requires no additional full-order solves. The process is repeated until a prescribed maximum number of training samples is reached.

Algorithm 3 Adaptive sampling for the POD-GPR reduced-order model

- 1: **Input:** Initial design $\mathcal{P}_M = \{\boldsymbol{\mu}_i\}_{i=1}^M$, candidate pool $\mathcal{C}$, maximum budget $M_{\max}$
 - 2: Compute the full-order snapshots $\{\mathbf{u}(\boldsymbol{\mu}_i)\}_{i=1}^M$ and assemble the snapshot matrix $\mathbf{X}$
 - 3: **while** $M < M_{\max}$ **do**
 - 4: Perform SVD on $\mathbf{X}$ to obtain the reduced basis Φ_r
 - 5: Project the snapshots to obtain $\mathbf{Z} = \mathbf{X}^\top \Phi_r \in \mathbb{R}^{M \times r}$
 - 6: Train the multi-output GPR with a shared kernel on $\mathcal{D}_M = \{(\boldsymbol{\mu}_i, \mathbf{a}(\boldsymbol{\mu}_i))\}_{i=1}^M$
 - 7: Evaluate the uncertainty indicator $\xi_{\text{sd}}(\boldsymbol{\mu}_j) = \hat{\sigma}(\boldsymbol{\mu}_j)$ for all $\boldsymbol{\mu}_j \in \mathcal{C}$
 - 8: $\boldsymbol{\mu}_{M+1} \leftarrow \arg \max_{\boldsymbol{\mu} \in \mathcal{C}} \xi_{\text{sd}}(\boldsymbol{\mu})$
 - 9: Perform a full-order simulation at $\boldsymbol{\mu}_{M+1}$ and append $\mathbf{u}(\boldsymbol{\mu}_{M+1})$ to $\mathbf{X}$
 - 10: Remove the selected point, $\mathcal{C} \leftarrow \mathcal{C} \setminus \{\boldsymbol{\mu}_{M+1}\}$
 - 11: $M \leftarrow M + 1$
 - 12: **end while**
 - 13: **Output:** Reduced basis Φ_r and trained GPR surrogate
-

By selectively performing high-fidelity simulations at the parameter locations where the surrogate is least certain, the adaptive procedure (Algorithm 3) allocates computational resources more efficiently than purely space-filling strategies such as Latin hypercube sampling and attains accurate reduced-order models with fewer full-order simulations. The performance of this workflow is assessed in Section 4.1.2.

4. Numerical results

This section evaluates the performance of the proposed reduced-order modeling (ROM) framework for the viscoelastic extrudate-swell problem. The numerical experiments are organized to progressively increase the complexity of the parameter space. We begin with a steady-state planar extrudate-swell configuration in which the relaxation time λ is varied. The parameter space is then extended to two dimensions by considering simultaneous variations of the relaxation time λ and the viscosity ratio $\beta = \eta_s/(\eta_s + \eta_p)$. We also investigate the effectiveness of the proposed adaptive sampling strategy in selecting informative training parameters and reducing the number of full-order model (FOM) simulations required to achieve a desired level of accuracy. Next, the framework is applied to a time-dependent scenario in which the solution depends on the parameter set (λ, β, t) , allowing us to assess the ability of the ROM to capture temporal dynamics in

addition to parametric variations. Finally, the methodology is extended to the three-dimensional cylindrical extrudate-swell configuration, where the parameter space again consists of (λ, β) .

The accuracy of the ROM is quantified using the relative ℓ^2 error of the predicted solution field at each test parameter value:

$$\varepsilon_u(\boldsymbol{\mu}) = \frac{\|\mathbf{u}_{\text{rom}}(\boldsymbol{\mu}) - \mathbf{u}_{\text{fom}}(\boldsymbol{\mu})\|_2}{\|\mathbf{u}_{\text{fom}}(\boldsymbol{\mu})\|_2}, \quad \text{and} \quad \varepsilon_c(\boldsymbol{\mu}) = \frac{\|\text{tr}(\mathbf{c})_{\text{rom}}(\boldsymbol{\mu}) - \text{tr}(\mathbf{c})_{\text{fom}}(\boldsymbol{\mu})\|_2}{\|\text{tr}(\mathbf{c})_{\text{fom}}(\boldsymbol{\mu})\|_2}, \quad (35)$$

where $\text{tr}(\mathbf{c})$ denotes the discrete field of the conformation-tensor trace. Here $\|\cdot\|_2$ denotes the Euclidean norm over all degrees of freedom. For a test set $\mathcal{T} = \{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_{M_{\text{test}}}\}$, we report the *mean* relative errors as

$$\bar{\varepsilon}_u(\mathcal{T}) = \frac{1}{M_{\text{test}}} \sum_{i=1}^{M_{\text{test}}} \varepsilon_u(\boldsymbol{\mu}_i), \quad \text{and} \quad \bar{\varepsilon}_c(\mathcal{T}) = \frac{1}{M_{\text{test}}} \sum_{i=1}^{M_{\text{test}}} \varepsilon_c(\boldsymbol{\mu}_i). \quad (36)$$

4.1. 2D planar extrudate swell

In this section, we evaluate the performance of the ROM for the planar extrudate-swell problem. The quantities of interest are the velocity field and the conformation tensor, which describe the flow kinematics and the viscoelastic stress state of the fluid, respectively. The assessment is carried out for both steady-state and transient solutions. For the steady-state problem, each high-fidelity simulation is advanced until a fully developed swollen configuration is obtained, while for the transient problem, the evolution of the flow toward this final state is considered. In addition, we investigate the performance of adaptive sampling on the ROM accuracy by comparing adaptive sampling with Latin hypercube sampling (LHS).

4.1.1. Steady-state solutions

The problem is characterized by three dimensionless numbers: the Weissenberg number $\text{Wi} = \lambda U_{\text{avg}}/H$, the viscosity ratio β , and the mobility parameter α . Throughout this study, the mobility parameter is fixed at $\alpha = 0.2$. Since U_{avg} and H are kept constant, variations in the relaxation time λ directly correspond to variations in the Weissenberg number. Before evaluating the accuracy of the ROM, we first examine how the rheological parameters (λ, β) influence the underlying flow fields that must be represented by the reduced basis. Both parameters affect the elastic stresses responsible for extrudate swell, although through different mechanisms. The relaxation time λ controls the Weissenberg number and determines the rate at which stresses accumulated inside the die relax after the fluid exits. Increasing λ results in increased swelling, as illustrated in Figure 2. In contrast, the viscosity ratio β controls the relative contribution of the polymeric stress to the total stress. As β approaches unity, the polymeric contribution decreases, and the resulting extrudate swell becomes progressively weaker, regardless of the value of λ .

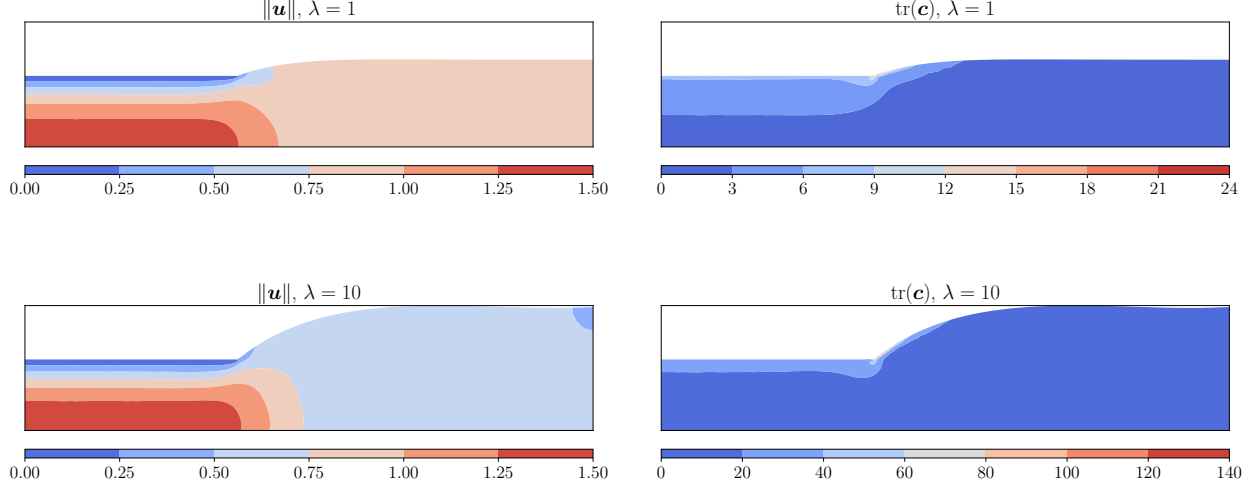


Figure 2: Influence of the relaxation time $\lambda = 1$ (top) and $\lambda = 10$ (bottom) on velocity magnitude and trace of the conformation tensor.

We first consider a one-parameter problem in which the relaxation time is varied over $\lambda \in [1, 10]$, while the viscosity ratio is held fixed at $\beta = 0.2$. The snapshot matrix is assembled from 10 high-fidelity solutions uniformly distributed over this interval, and the reduced basis follows from its truncated SVD. Projecting the snapshots onto this basis yields the reduced coefficients shown in Figure 3. The ROM is evaluated on an independent test set of $M_{\text{test}} = 100$ parameter values, generated by Latin hypercube sampling within the same training domain.

Figure 4 presents the normalized singular values associated with the velocity field, the trace of the conformation tensor, and the mesh deformation. All three exhibit a rapid decay, indicating that the solution manifold generated by variations in λ can be represented effectively in a low-dimensional space. The singular values of the mesh deformation decrease most rapidly, followed by those of the velocity field, whereas those of the conformation trace decay more slowly.

The spatial distributions of the relative error for the representative values $\lambda = 1.5$ and $\lambda = 9.5$ are shown in Figure 5. In both cases, the ROM reproduces the corresponding full-order solution accurately over most of the computational domain. The largest discrepancies are localized near the die exit, where the polymeric stresses vary sharply. Consistent with the stronger elastic response associated with larger relaxation times, the errors for $\lambda = 9.5$ are slightly higher than those observed for $\lambda = 1.5$.

Finally, Figure 6 summarizes the convergence of the ROM error as the number of retained POD modes is increased. The mean relative error initially decreases rapidly and subsequently reaches a plateau of approximately 1.3×10^{-5} for the velocity field and 1.7×10^{-4} for the trace of the conformation tensor. The

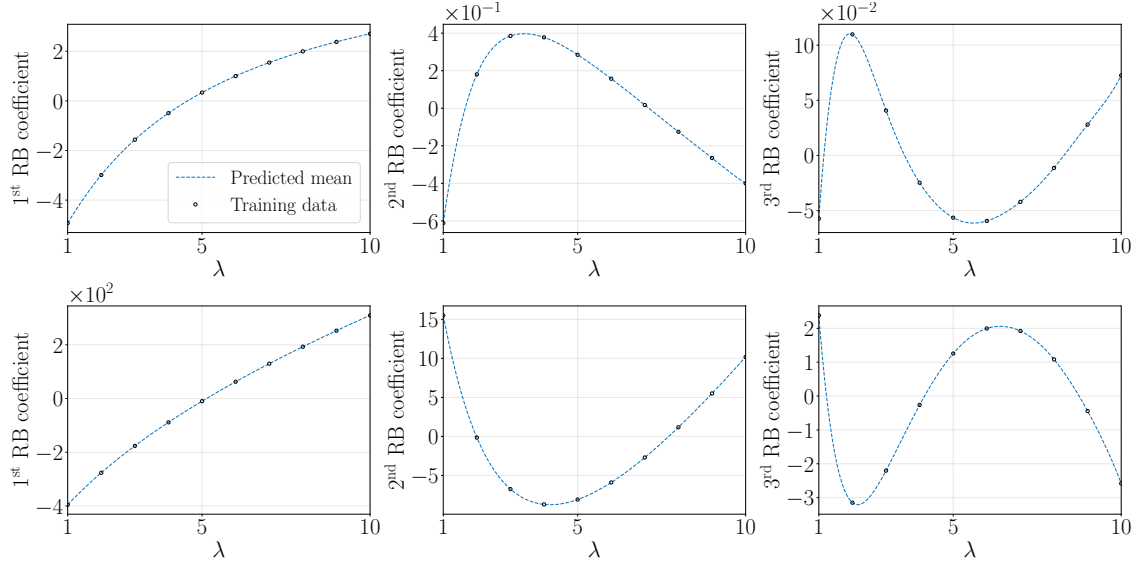


Figure 3: The first three POD coefficients for the velocity field (top) and the trace of the conformation tensor (bottom).

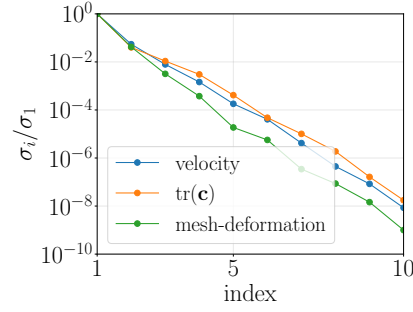


Figure 4: Normalized singular values for velocity, trace of the conformation tensor, and mesh-deformation fields.

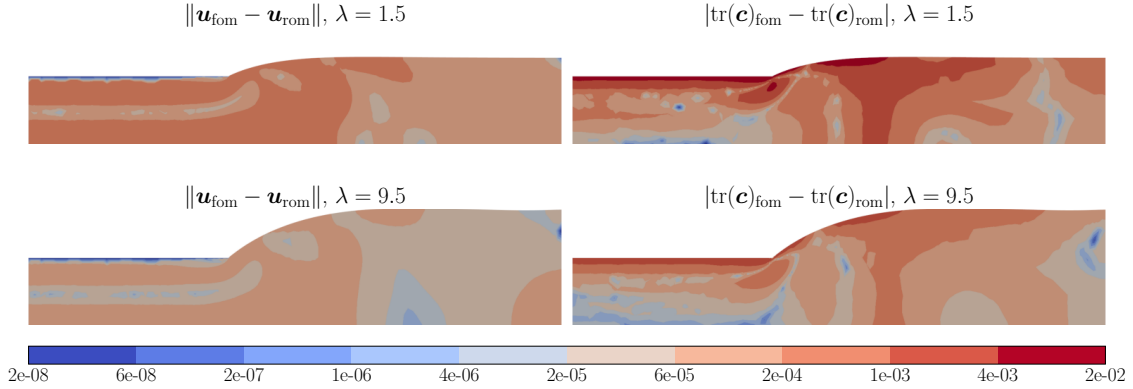


Figure 5: Spatial distributions of the reconstruction error for velocity and trace of conformation tensor fields at $\lambda = 1.5$ (top) and $\lambda = 9.5$ (bottom).

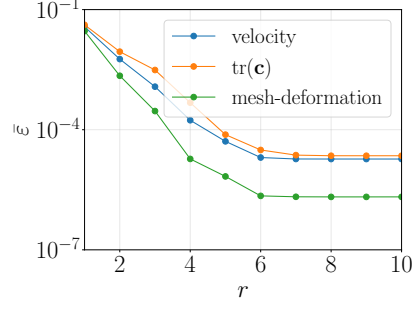


Figure 6: Convergence analysis of the POD-GPR reduced-order model.

results indicate that approximately five to eight POD modes are sufficient to capture the dominant structures of the solution manifold. Beyond this range, the contribution of the POD truncation error becomes small, and the overall ROM accuracy is primarily limited by the GPR approximation used to map the relaxation time to the corresponding reduced coefficients.

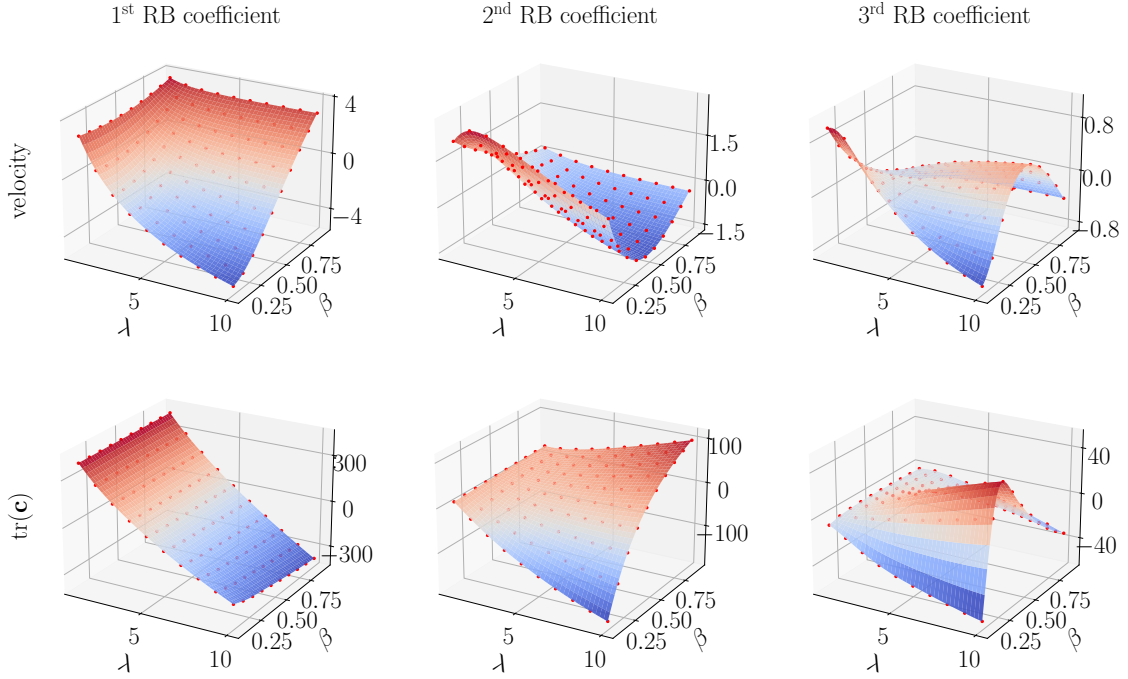


Figure 7: The first three reduced coefficients for the velocity field (top) and the trace of the conformation tensor field (bottom).

We now consider a more challenging problem in which both rheological parameters (λ, β) vary simultaneously. The resulting two-dimensional parameter domain is defined as

$$\lambda \in [1, 10], \quad \beta \in [0.1, 0.9].$$

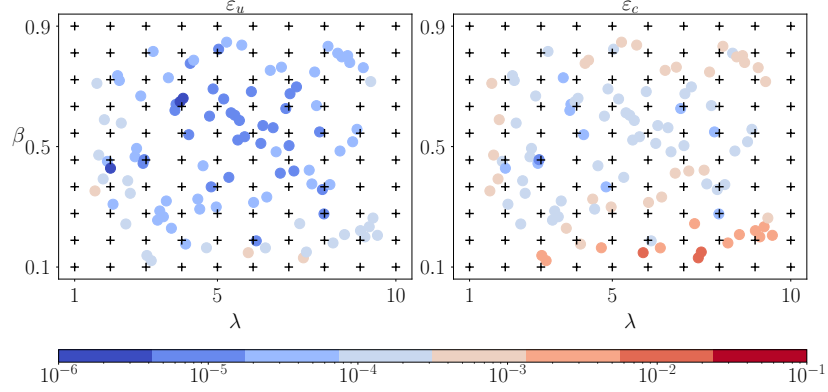


Figure 8: Point-wise relative errors for the velocity (ε_u) and the trace of the conformation tensor (ε_c) over the (λ, β) parameter space. Plus markers (+) indicate the training samples and the filled circles are the test parameters, colored by their relative error.

The training set is built on a tensor-product grid with 10 uniformly spaced values in each parameter direction, giving 100 high-fidelity snapshots. Representative POD coefficient surfaces for the velocity and trace of conformation tensor fields are shown in Figure 7. Compared with the single-parameter case, simultaneous variation of λ and β introduces additional complexity into the solution manifold. Nevertheless, the dominant reduced coefficients remain smooth functions of the parameters, which supports their approximation using GPR. The predictive performance of the ROM is assessed at 100 test parameter points generated via Latin hypercube sampling (LHS) within the same parameter domain employed for training. These samples are used exclusively for ROM interpolatory validation and are chosen such that none of them coincide with any of the training locations. The point-wise relative errors for the velocity field and for the trace of the conformation tensor are reported in Figure 8. The resulting error distribution is non-uniform over the parameter domain, indicating that the accuracy of the surrogate approximation depends on the particular parameter combination. In particular, larger errors are observed more frequently in regions of lower β , especially for the conformation tensor, where stronger polymeric effects may enhance the sensitivity of the solution with respect to the model parameters.

It should be noted, however, that the local GPR error is governed not only by the intrinsic complexity of the parameter-to-solution map but also by the spatial distribution of the available training samples in parameter space. Consequently, regions of elevated error may arise from increased nonlinear parameter dependence, insufficient local sampling density, or a combination of these factors.

Finally, Figure 9 reports the convergence of the mean relative error with the number of retained POD modes. The mean error decreases as modes are added and reaches a plateau at roughly 20 modes for both fields. The trace of the conformation tensor has a higher mean error than the velocity, consistent with

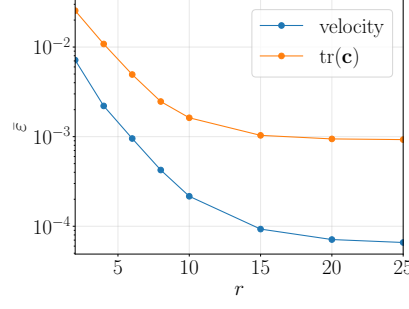


Figure 9: Convergence of the mean relative error with respect to the number of retained POD basis functions.

its slower singular-value decay. Therefore, around 20 POD modes suffice to represent the two-parameter solution manifold, even though the simultaneous variation of λ and β enriches the complexity relative to the single-parameter case.

4.1.2. Adaptive sampling

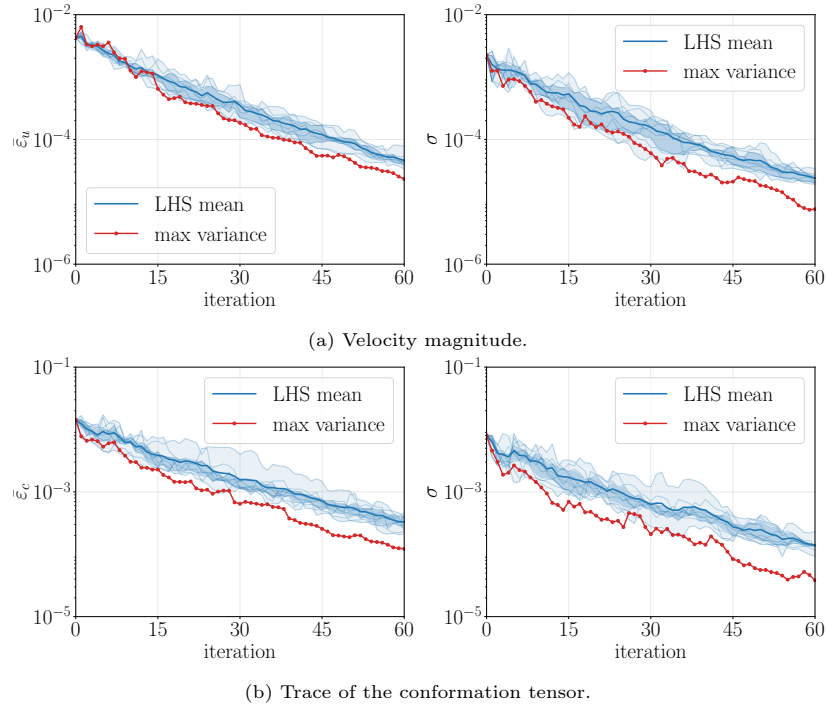


Figure 10: Comparison of adaptive sampling and multiple LHS designs for different seeds in the two-parameter ROM in (λ, β) .

The adaptive sampling strategy of Section 3.4 is compared with the classical space-filling LHS for the two-parameter case $(\lambda, \beta) \in [1, 20] \times [0.1, 0.9]$. Here, the relaxation-time range is deliberately extended to provide a more challenging setting where adaptive sampling can allocate new samples more efficiently as compared to LHS. In both strategies, we start from the same initial design of 10 LHS points and add one

snapshot per iteration, up to 60 iterations. At each iteration, the adaptive strategy selects from a fixed pool of 10,000 LHS candidates covering the parameter domain and chooses the sampling point that maximizes the GPR predictive standard deviation. Accuracy is evaluated after every iteration on an independent test set of 500 LHS points drawn from the same domain. Since an LHS design depends on its random seed, the space-filling LHS is repeated for 10 seeds.

Figure 10 reports the mean relative error (left column) and the maximum predictive standard deviation (right column) over the test set for the velocity magnitude (top row) and the trace of the conformation tensor (bottom row). Adaptive sampling is performed independently for the velocity and conformation-trace ROMs. Both quantities exhibit a clear overall downward trend as additional snapshots are added. The adaptive sampling strategy achieves lower reconstruction errors than the ten LHS realizations over most of the sampling range. A similar trend is observed for the predictive standard deviation, which decreases alongside the reconstruction error. Since the predictive standard deviation is obtained directly from the GPR model without requiring additional full-order simulations, it provides a computationally inexpensive measure of surrogate uncertainty. Enriching the training set at those locations places each new snapshot where it is most informative and, therefore, produces a faster decay compared to the LHS strategy.

4.1.3. Time-dependent solutions

The first two cases addressed steady-state solutions. We now extend the framework to the full transient evolution of the flow towards steady state by treating the simulation time t as an additional parameter, so that $\boldsymbol{\mu} = (\lambda, \beta, t)$. The training set combines five logarithmically spaced time instances in $t \in [0.1, 20]$ with the same 10×10 grid in (λ, β) , yielding 500 snapshots that span both the rheological parameters and the temporal dynamics. The logarithmic spacing reflects the fact that the flow evolves rapidly at early times and then relaxes slowly towards steady state, so that a uniform distribution in t would leave the initial transient under-resolved. The ROM is tested on a further 500 snapshots, obtained by combining 100 Latin hypercube samples in (λ, β) with the same five time instances used for training. The test set therefore probes interpolation in the rheological parameters at time levels that the reduced basis has already encountered, isolating the parametric error from any error associated with interpolation in time.

We study the convergence of the mean relative error with respect to the number of POD modes, as shown in Figure 12. The results indicate that approximately 40 modes are sufficient for the error to reach a plateau. This behavior demonstrates that, despite the increased complexity introduced by the transient dynamics, the solution manifold remains low-dimensional. Similarly, the spatial distribution of the relative errors for the velocity field and the trace of the conformation tensor field, averaged over all sampled time instances, is

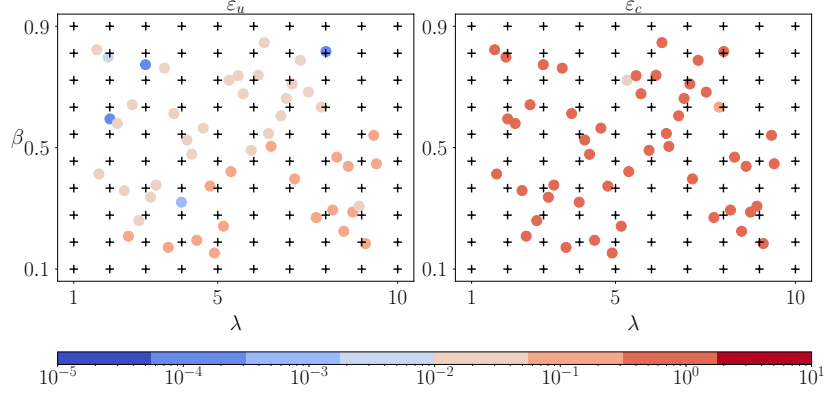


Figure 11: Point-wise relative errors over the (λ, β) parameter space, averaged over the sampled time instances. Plus markers (+) indicate the training samples and filled circles are the test parameters, colored by their relative error.

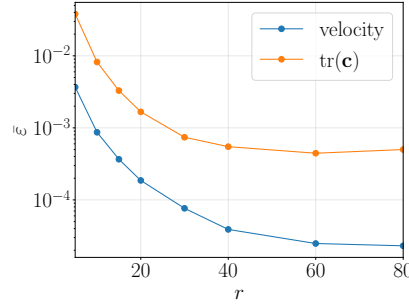


Figure 12: Convergence of the time-averaged mean relative error with respect to the number of retained POD modes.

presented in Figure 11. The point-wise velocity error is uniformly distributed throughout, but the point-wise error in the trace of the conformation tensor is largest near the region where the β is small.

4.2. Three-dimensional cylindrical extrudate swell

The results obtained for the planar two-dimensional configuration demonstrate that the POD-GPR framework provides an accurate approximation of the parametric extrudate-swell problem in both steady and transient regimes. We now extend the assessment to a three-dimensional cylindrical extrudate-swell configuration.

Figure 13 displays the FOM solutions for the velocity magnitude and the trace of the conformation tensor at two relaxation times, $\lambda = 1$ and $\lambda = 10$. As in the two-dimensional case, increasing the relaxation time produces a more pronounced swell.

To quantify the ROM accuracy in this setting, the training set is built on a tensor-product grid with 10 uniformly spaced values in each parameter direction over $(\lambda, \beta) \in [1, 10] \times [0.1, 0.9]$, giving 100 high-fidelity snapshots. The test set consists of 50 points generated by Latin hypercube sampling (LHS) within the same

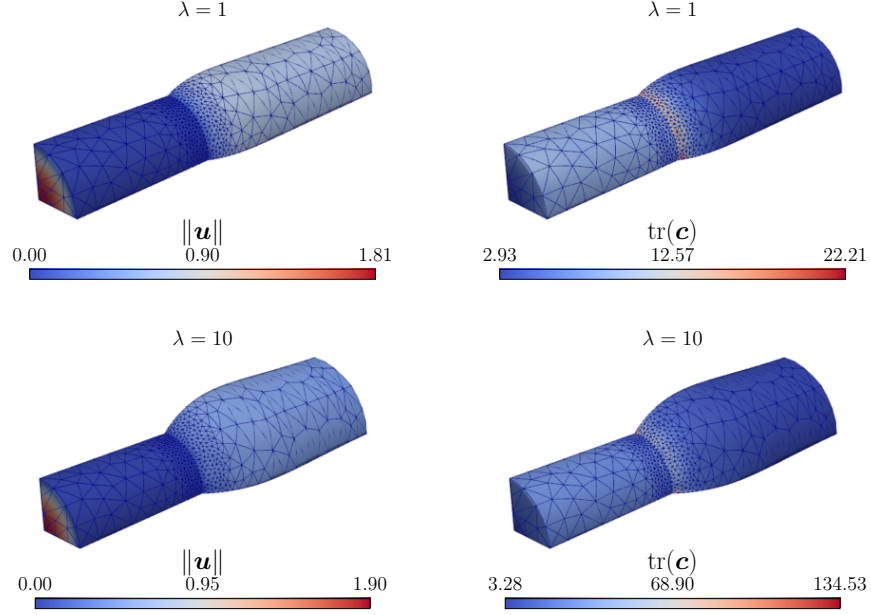


Figure 13: Velocity magnitude and trace of the conformation tensor for the three-dimensional extrudate-swell configuration at $\lambda = 1$ (top) and $\lambda = 10$ (bottom).

parameter domain, none of which coincide with a training sample. The convergence of the mean relative error as a function of the number of retained POD modes is reported in Figure 14. An analogous trend to the two-dimensional case is observed: the velocity field is approximated with systematically lower error than the trace of the conformation tensor. For approximately 20 modes, the error levels off, indicating that the inclusion of additional modes does not produce a substantial further reduction in the mean relative error.

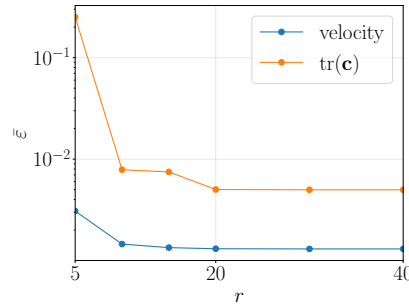


Figure 14: Convergence of the mean relative error as a function of the number of retained POD modes for the three-dimensional ROM over the parameter space (λ, β) .

The point-wise relative errors over the (λ, β) parameter space for the three-dimensional ROM are shown in Figure 15. The resulting error distribution closely resembles that of the two-dimensional configuration: the ROM delivers accurate predictions over most of the parameter domain, while larger errors are localized

in regions of low β . The same physical mechanisms identified for the two-dimensional case therefore appear to control the three-dimensional response, supporting the consistency and robustness of the POD-GPR framework across spatial dimensions.

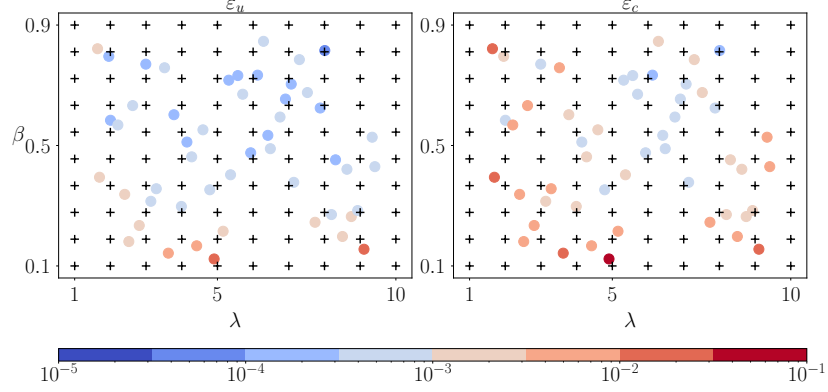


Figure 15: Point-wise relative errors for the velocity (ε_u) and the trace of the conformation tensor (ε_c) in the three-dimensional configuration. Plus markers (+) indicate the training samples and filled circles are the test parameters, colored by their relative error.

Figure 16 illustrates the spatial distribution of the ROM reconstruction errors at two representative parameter points, $(\lambda, \beta) = (2, 0.15)$ and $(9.5, 0.15)$, which are representative of mildly and strongly elastic regimes, respectively. As in the two-dimensional configuration, the velocity field is reconstructed with high accuracy throughout the domain. In contrast, the trace of conformation tensor field exhibits larger discrepancies, primarily concentrated in the vicinity of the die exit, and these discrepancies intensify with increasing relaxation time.

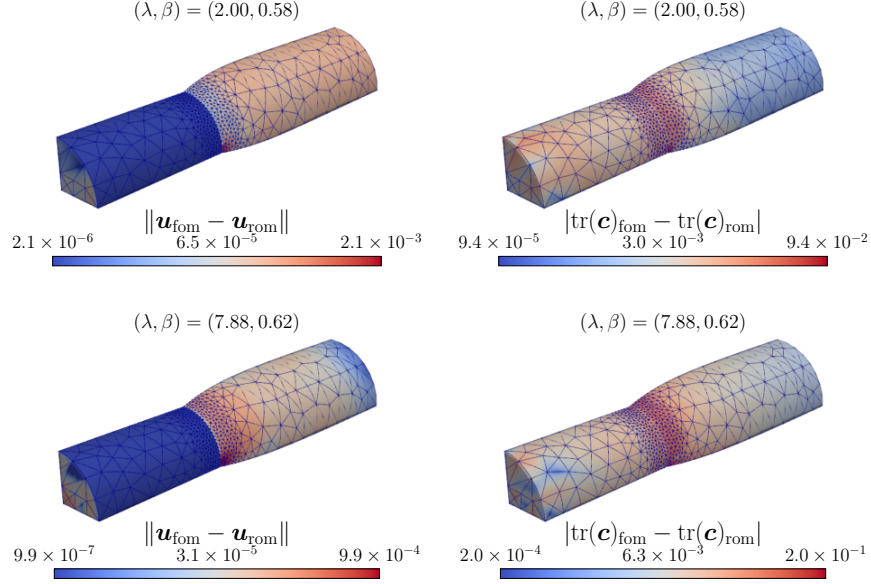


Figure 16: Spatial distribution of ROM reconstruction errors for the velocity and trace of conformation tensor fields at two parameter points in the three-dimensional configuration.

5. Conclusions

This work presented a non-intrusive reduced-order modeling framework for viscoelastic extrudate-swell flows based on proper orthogonal decomposition (POD) and Gaussian process regression (GPR). The approach treats the high-fidelity finite-element solver as a black box and learns the dependence of the reduced coefficients on the rheological parameters directly from simulation data. Separate reduced representations are constructed for the velocity field, the trace of the conformation tensor, and the deforming mesh coordinates, allowing the flow fields and the swollen geometry to be reconstructed at new parameter values without further calls to the full-order solver. The framework was assessed for both two-dimensional planar and three-dimensional cylindrical extrudate-swell configurations governed by the Giesekus model.

The numerical results show that the solution manifolds associated with the investigated parameter ranges admit compact POD representations. For the one-parameter planar problem, in which only the relaxation time λ is varied, approximately 5–8 modes are sufficient for the reconstruction error to reach its plateau. When both λ and the viscosity ratio β are varied, approximately 20 modes are required, while inclusion of time as an additional parameter increases this number to approximately 40. Among the reconstructed quantities, the mesh coordinates exhibit the fastest singular-value decay, followed by the velocity field, whereas the trace of the conformation tensor requires a larger reduced space. This behavior is consistent with the sharper spatial variations of the viscoelastic stresses, particularly in the vicinity of the die exit.

For the one-parameter steady problem, the mean relative errors reach approximately 1.3×10^{-5} for the velocity and 1.7×10^{-4} for the trace of the conformation tensor. Increasing the dimension of the parameter space leads to larger interpolation errors, as expected, but the reduced model remains accurate throughout most of the investigated parameter domains. The largest errors are generally observed for the conformation-trace field and are concentrated toward smaller values of β , where the polymeric contribution to the total stress is stronger. The same qualitative behavior is observed for the three-dimensional cylindrical configuration, demonstrating that the proposed POD–GPR methodology can be applied without modification when moving from the planar to the three-dimensional extrudate-swell problem.

An uncertainty-driven adaptive sampling strategy was additionally developed by using the GPR posterior standard deviation as the sampling indicator. Starting from an initial Latin hypercube design, new full-order simulations are placed sequentially at parameter locations of maximum predictive uncertainty. For the two-parameter problem over $(\lambda, \beta) \in [1, 20] \times [0.1, 0.9]$, this strategy produces lower mean reconstruction errors than the average of the Latin hypercube designs over most of the enrichment process for both the velocity and conformation-trace ROMs. The simultaneous decrease of the predictive standard deviation and the observed ROM error also indicates that the GPR uncertainty provides a useful and inexpensive indicator for directing additional high-fidelity simulations.

The computational results further illustrate the motivation for the non-intrusive formulation. A single full-order simulation requires approximately 28 s for the planar problem and 718 s for the cylindrical problem, whereas an online POD–GPR evaluation requires approximately 0.49 ms and 1.37 ms, respectively. Relative to the corresponding transient full-order simulations used to obtain the fully swollen state, these timings correspond to speedups of approximately 5×10^4 in two dimensions and 5×10^5 in three dimensions. The additional offline costs associated with POD construction and GPR training are small compared with the cost of generating the high-fidelity training snapshots. Consequently, the proposed framework is particularly attractive in multi-query settings in which the offline cost can be amortized over a large number of parameter evaluations.

The present study also identifies several directions for further development. In the transient experiments, the ROM was evaluated at the same discrete time levels used in the training set; validation of interpolation at previously unseen time instances therefore remains an important extension. Increasing the parameter dimension to include quantities such as the Giesekus mobility parameter α , flow rate, and die geometry would provide a more demanding test of both the reduced representation and the adaptive sampling procedure. Finally, the low online cost of the present ROM makes it suitable for integration with computationally

intensive multi-query applications such as process optimization, uncertainty quantification, and Bayesian parameter inference.

Appendix A. Mesh convergence studies

Before constructing the ROM, we verify the FOM accuracy through a mesh-convergence study to ensure that the snapshots used for basis construction are reliable. Starting from a coarse discretization $M1$, successive uniform refinements are applied up to mesh $M4$, which serves as the reference solution. The relative errors in velocity magnitude u_h and conformation-tensor trace c_h along the sampling line Γ_s shown in Figure A.17, located at $y = 0.5$, are

$$e_u = \frac{\left(\int_{\Gamma_s} \|u_h - u^*\|^2 ds \right)^{1/2}}{\left(\int_{\Gamma_s} \|u^*\|^2 ds \right)^{1/2}}, \quad e_c = \frac{\left(\int_{\Gamma_s} (c_h - c^*)^2 ds \right)^{1/2}}{\left(\int_{\Gamma_s} (c^*)^2 ds \right)^{1/2}},$$

where u^* and c^* denote the corresponding fields of the reference solution $M4$. The sampling line (Figure A.17), the profiles (Figure A.18), and the convergence rates for meshes $M1$ - $M4$ (Figure ??) show that the mesh $M2$ provides a good balance of accuracy and computational cost; it is therefore used for all remaining 2D FOM simulations.

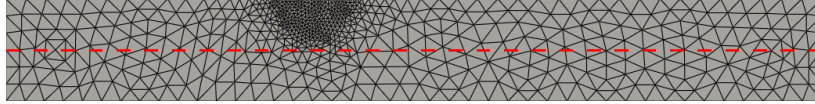


Figure A.17: Mesh $M2$ used in the mesh-convergence study. The sampling line for velocity and trace of the conformation tensor evaluation is shown in dashed red.

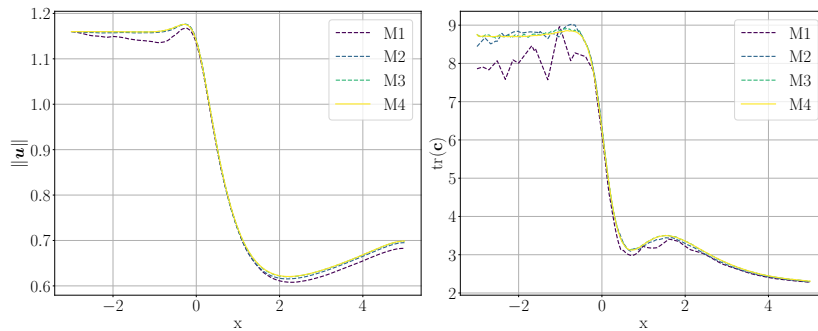


Figure A.18: Sampled profiles along Γ_s for the planar extrudate-swell problem: velocity magnitude (left) and trace of the conformation tensor (right).

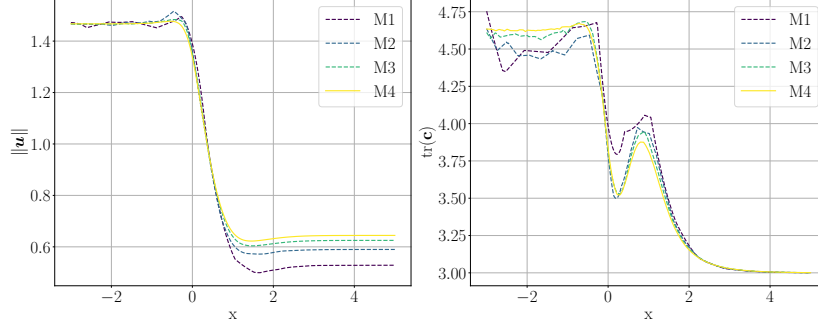


Figure A.19: Sampled profiles along Γ_s for the planar extrudate-swell problem: velocity magnitude (left) and trace of the conformation tensor (right).

Appendix B. Computational cost

Table B.1 summarizes the offline and online costs of the reduced-order model for the two-parameter planar and cylindrical configurations. All computations were performed on an ASUS Vivobook laptop equipped with an AMD Ryzen 7 7730U processor (8 cores, 16 threads, 2.0 GHz) and 16 GB of RAM, running Windows 11. Full-order simulations were carried out using the finite-element solver described in Section 2, while the reduced-order model was implemented in Python 3.14.4 using NumPy 2.4.4 and SciPy 1.17.1. Online timings are reported as the mean over 200 repetitions of a single parameter query. The offline stage is dominated by the generation of the training snapshots. A single full-order simulation requires 28 s for the planar configuration and 718 s for the cylindrical configuration, corresponding to 200 time steps with $\Delta t = 0.1$ in both cases. By comparison, the truncated SVD and Gaussian-process training require only 13.7 s and 16.9 s, respectively, accounting for less than 0.5% of the total offline cost. Once trained, the reduced-order model predicts the steady-state velocity field in 0.49 ms for the planar case and 1.37 ms for the cylindrical case. Relative to the corresponding transient full-order simulations, this yields speedups of approximately 5×10^4 and 5×10^5 , respectively. These speedups compare a direct reduced-order prediction of the steady-state solution with a transient full-order simulation evolved to the fully swollen state. Note that the FOM solver can be optimized further by *e.g.* a direct calculation of the steady state using implicit methods.

case	M	N_h	modes r_u	<i>offline</i>		<i>online</i>
				FOM [s]	SVD + GPR (training) [s]	GPR (evaluation)[ms]
2D planar, (λ, β)	100	4690	42	28	13.7	0.49
3D cylindrical, (λ, β)	100	23124	51	718	16.9	1.37

Table B.1: Offline and online costs of the POD-GPR reduced-order model. Here, N_h denotes the number of velocity degrees of freedom and r_u the number of retained velocity modes. The FOM column reports the cost of a single full-order simulation, implying a total snapshot-generation cost of M times this value. The “SVD + GPR” column reports the cost of basis construction and surrogate training. Online timings correspond to the median of 200 single-parameter queries.

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